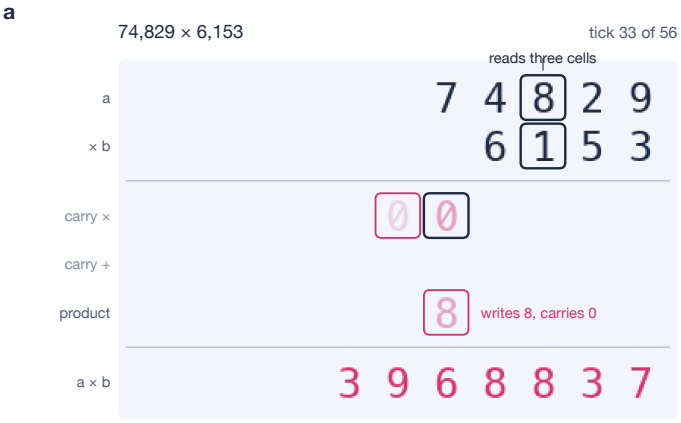
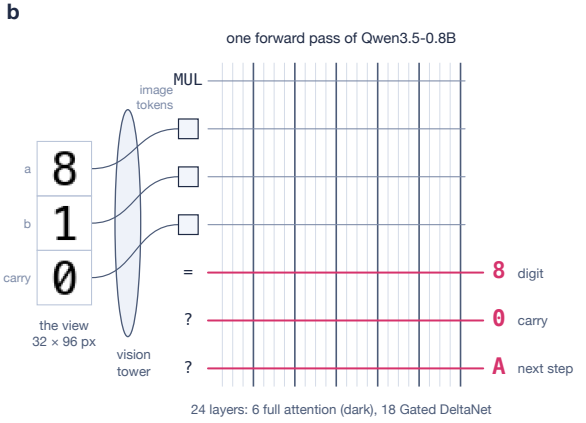
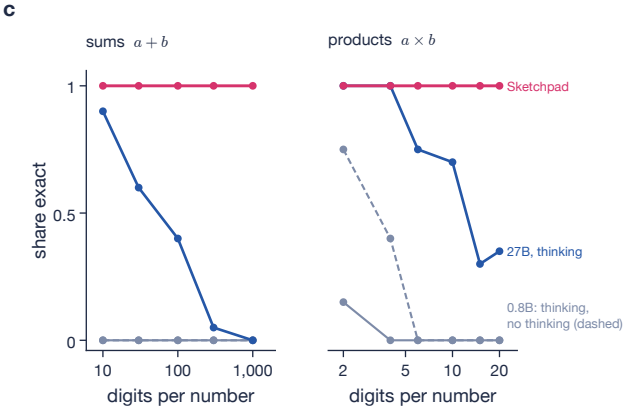
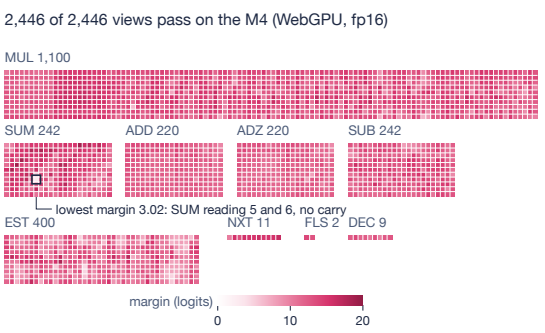

a
74,829 × 6,153
tick 33 of 56
reads three cells
a
× b
7 4 8 2 9
6 1 5 3
carry ×
carry +
product
writes 8, carries 0
a × b
3 9 6 8 8 3 7


b
one forward pass of Qwen3.5-0.8B
image tokens
MUL
a
b
carry
8
1
0
the view
32 × 96 px
vision tower
=
?
?
8 digit
0 carry
A next step
24 layers: 6 full attention (dark), 18 Gated DeltaNet


c
sums $a + b$
products $a \times b$
share exact
1
0.5
0
10
100
1,000
digits per number
Sketchpad
27B, thinking
0.8B: thinking,
no thinking (dashed)
2
5
10
20
digits per number


d
2,446 of 2,446 views pass on the M4 (WebGPU, fp16)
MUL 1,100
SUM 242
ADD 220
ADZ 220
SUB 242
lowest margin 3.02: SUM reading 5 and 6, no carry
EST 400
NXT 11
FLS 2
DEC 9
margin (logits)
0
10
20